

Sanaz Rabinia

SUMMARY

I received my Ph.D. in Computer Science from Wayne State University, specializing in algorithm design, combinatorial optimization, and applied machine learning/AI. I develop and design innovative algorithms for Edge Computing systems, large-scale graph problems, and real-world predictive modeling applications. I was recognized with the 2024 NCWIT AiC National Collegiate Award–Honorable Mention from the National Center for Women & Information Technology, the 2025 Ralph H. Kummeler Distinguished Achievement Award from Wayne State University, and the 2026 IEEE TCSC Outstanding Ph.D. Dissertation Award from The IEEE Technical Committee on Scalable Computing (TCSC) for impactful contributions to Algorithms and Edge Computing systems.

RESEARCH INTERESTS

Edge Computing, Cloud Computing, Parallel Algorithms, Randomized Algorithms, Approximation Algorithms, Applied Machine Learning, Game Theory

EDUCATION

- 1/2019 - 7/2025 **Ph.D., Computer Science**
Department of Computer Science,
Wayne State University, Detroit, MI
Minor: Mathematics
Advisor: Dr. Daniel Grosu
- 9/2011 - 9/2013 **M.Sc., Mathematics, Sharif University of Technology, Iran**
- 9/2007 - 7/2011 **B.Sc., Mathematics, Shahid Beheshti University, Iran**

PUBLICATIONS

Journal Articles

- J1.** Speeding-up Graph Algorithms via Parallel Randomized Clique Partitioning
A. Chavan, S. Rabinia, M. Brocanelli, D. Grosu, submitted to **Transactions on Parallel and Distributed Systems**, 2025.
- J2.** Online Algorithms for Data Sharing-Aware Task Allocation in Edge Computing Systems
S. Rabinia, D. Grosu, **IEEE Transactions on Cloud Computing**, Vol. 14, No. 3, pp. 1989-2004, September 2026.
- J3.** Speeding-up Graph Algorithms via Clique Partitioning
A. Chavan, S. Rabinia, M. Brocanelli, D. Grosu, **Networks**, Vol. 88, No. 1, pp. 85–98, July 2026.
- J4.** Algorithms for Data Sharing-Aware Task Allocation in Edge Computing Systems
S. Rabinia, N. Didar, D. Grosu, M. Brocanelli, **IEEE Transactions on Parallel and Distributed Systems**, Vol. 36, No. 1, pp. 15-28, January 2025.
- J5.** Machine Learning Approach to Forecast Work Zone Mobility using Probe Vehicle Data
M. Kamyab, S. Remias, E. Najmi, S. Rabinia, JM. Waddell, **Transportation Research Record**, SAGE Publications, Vol. 2674, No. 9, pp. 157-167, July 12, 2020.

Conference Papers

- C1.** Data Sharing-Aware Online Algorithms for Task Allocation in Edge Computing Systems
S. Rabinia, D. Grosu, **Proc. of the IEEE International Conference on Edge Computing (IEEE EDGE 2024)**, July 7-13, 2024, Guangzhou, China.
- C2.** Data Sharing-Aware Algorithms for Task Allocation in Edge Computing Systems
S. Rabinia, D. Grosu, **Proc. of the IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid 2024)**, May 6-9, 2024, Philadelphia, USA.
- C3.** Data Sharing-Aware Task Allocation in Edge Computing Systems
S. Rabinia, H. Mehryar, M. Brocanelli, D. Grosu, **Proc. of the IEEE International Conference on Edge Computing (IEEE EDGE 2021)**, December 18-20, 2021, Guangzhou, China.

Book Chapter

- B1.** A Comprehensive Study of Game Theory Applications for Smart Grids and Demand Side Management Programs, and Transportation Networks.
A. Mohammadi, S. Rabinia, **Smart Microgrids**, pp 57-64, Springer, January 18, 2019.

FELLOWSHIPS, AWARDS AND HONORS

- Received the 2026 IEEE TCSC (Technical Committee on Scalable Computing) Outstanding Ph.D. Dissertation Award, Technical Committee on Scalable Computing, October 2026.
- Received the 2025 Ralph H. Kummeler Distinguished Achievement Award in Graduate Student Research, Wayne State University, April 2025. ([Link to Award](#))
- Panelist of Michigan Council of Women in Technology Foundation Conference (MICWIC), October 2024.
- Received the 2024 travel award from the National Center for Women and Information Technology (NCWIT), October 2024.
- Received the 2024 Association for Computing Machinery (ACM) Certificate as top 5 posters presented at Tapia2024 for my poster with title: "Data Sharing-Aware Online Algorithms for Task Allocation in Edge Computing Systems", September 2024.
- The Graduate Student Professional Travel Award from the department of Computer Science, Wayne State University, April and June 2024.
- Finalist in the Doctoral Consortium track to present my doctoral dissertation at the CMD-IT/Tapia Conference 2024.
- Received the 2024 NCWIT AiC Collegiate Award - Honorable Mention and Monetary Prize from the National Center for Women & Information Technology (NCWIT), May 2024. ([Link to Award](#))
- Full Scholarship to attend the 2024 NCWIT Summit, May 2024.
- Full Scholarship to attend the CMD-IT/Tapia Conference 2024, May 2024.
- Full Scholarship to attend the 24th IEEE/ACM International Symposium on Cluster, Cloud, and Internet Computing (CCGrid 2024), May 2024.
- Finalist of the 2024 National Center for Women & Information Technology (NCWIT) AiC Collegiate Award, January 2024.
- Graduate Research Excellence Award in Computer Science, Wayne State University, 2022.
- Full Scholarship to attend Grace Hopper Conference, September 21-23, 2022.
- Full Scholarship to attend CRA-WP Grad Cohort, Computing Research Association Women Workshop, April 22-24, 2021 & 2022.
- Outstanding Graduate Teaching Assistant Award in Computer Science, Wayne State University, 2020.

- Travel Award to attend the International Conference on Machine Learning (ICML), June 9-15, 2019.
- Travel Award to attend StringBio Conference at University of Central Florida, October 25-27, 2018.
- Travel Award to attend the Approximation Theory and Machine Learning Conference at Purdue University, September 29-30, 2018.

PROFESSIONAL EXPERIENCE

Algorithms (Sequential, Parallel, Randomized, & Approximation) Mar 2019 - Jul 2025

- Designed offline algorithms for task allocation in edge computing systems that take into account the sharing of data among tasks offloaded to the same server which led to up to 14 times less network load. (**Java**) ([Link to GitHub](#))
- Designed several heuristic online algorithms and a local search algorithm for task allocation in edge computing systems, considering the sharing of data among tasks offloaded to the same server, which led to up to 92% less data transferred over the network and up to 82% fewer servers utilized. (**Java**) ([Link to GitHub](#))
- Designed graph compression algorithms (sequential, parallel, randomized) based on bipartite clique partitioning, achieving $105\times$ faster execution and 26% higher compression compared to the best known algorithm. (**C, MPI**) ([Link to GitHub](#))

Optimization

Jan 2020 - Jul 2025

- Created a novel analytical model for capturing the sharing among offline tasks and formulated the problem as a bi-objective mixed-integer multilinear program, balancing profit maximization with resource/network minimization.
- Defined a novel mathematical model for capturing the sharing among online tasks, resulting in reducing the traffic in the edge network, the memory usage at the edge servers, and the number of servers needed to execute the tasks.

High Performance Computing & Graph Networks

Sep 2023 - Jul 2025

- Generated and analyzed billion-node bipartite graphs using Erdős-Rényi random graph model on Wayne State University supercomputer (Grid). (**R, Python**) ([Link to GitHub](#))
- Benchmarked parallel graph compression algorithms on 128-core Pittsburgh Supercomputing Center (Bridges-2), validating scalability and efficiency. (**MPI, OpenMP**) ([Link to GitHub](#))

Machine Learning, Deep Learning, & Artificial Intelligence

Jan 2020 - Apr 2025

- Built machine learning predictive models (Random Forest, XGBoost, ANN) for spatio-temporal forecasting of lane closures, improving prediction accuracy to 85%. (**ANN, XGBoost, Python**)
- Designed and trained convolutional neural networks (CNNs) for handwriting-author matching in bank check authentication and forensics, achieving 88% accuracy. (**CNN, Python**) ([Link to GitHub](#))
- Developed and assessed machine learning and statistical models that enhanced system performance, optimized data management, and increased predictive accuracy. (**Python**) ([Link to GitHub](#))

Natural Language Processing & Large Language Models

Jul 2024 - Dec 2025

- Developed and deployed conversational AI applications including RAG-based PDF Q&A, tool-augmented agents (Wikipedia, arXiv), NLP SQL querying, and summarization of YouTube and web content. (**Python, LangChain**) ([Link to GitHub](#))
- Developed a new search engine aimed at assisting analysis in conducting comprehensive search from Wikipedia and arXiv. (**Python, LangChain**) ([Link to GitHub](#))
- Developed and deployed a portfolio of LLM-driven applications using OpenAI API and Hugging Face showcasing end-to-end generative AI solutions; deployed apps run publicly on Streamlit Cloud. (**Python, LangChain**) ([Link to GitHub](#))

- Developed and deployed customer churn prediction models using deep learning for churn binary classification and regression, including hyperparameter tuning and Streamlit web applications. (**Python, LangChain**) ([Link to GitHub](#))
- Built a sentiment-based recommendation system using a recurrent neural network (RNN) trained on the IMDB movie review dataset and deployed it as a real-time Streamlit application. (**Python, LangChain**) ([Link to GitHub](#))
- Implemented language modeling for next-word prediction using LSTM/GRU networks, covering text preprocessing, tokenization, embedding, model training, and Streamlit web applications. (**Python, LangChain**) ([Link to GitHub](#))

COMPUTER SKILLS

- Supercomputer: Utilizing Wayne State University's supercomputer (Grid), Pittsburgh Supercomputing Center (Bridge-2)
- Programming Languages: C, C++, Java, Python, R, Qiskit
- Programming Tools: gcc, Visual Studio, IntelliJ IDEA, Spring boot, Hibernate
- Parallel Programming: CUDA, MPI, OpenMP, pThreads
- Machine Learning: Python, Octave, PySpark
- Deep Learning/AI: PyTorch, TensorFlow, LLM
- Machine Learning Frameworks: Hugging Face, Transformers
- Web Technology: JavaScript, HTML, CSS, PHP
- Databases: SQL Server, MySQL, AWS
- Software Engineering Methods: UML, Agile, REST
- Unit Testing: JUnit, PHPUnit, Laravel
- Other Tools: GitHub, Linux, Ubuntu, LaTeX, Corel, Continuous Integration

COURSEWORK AND ONLINE CERTIFICATE

University Coursework: Algorithm Design, Parallel Computing 1&2, Machine Learning, Distributed Systems, Data Mining, Database Management System, Advanced Algebra, Advanced Graph Theory, Real Analysis, Statistics 1 &2

Online Certified Coursework: Generative AI, Machine Learning, Artificial Intelligence, Large Language Models, Natural Language Processing, PySpark, AWS Cloud Practitioner, Qiskit Programming

Other Certificates: AWS Certified Cloud Practitioner, ACM Certified Reviewer

PRESENTATIONS

Conferences

- **Data Sharing-Aware Task Allocation Algorithms in EC Systems**
ACM CMD-IT/Tapia Conference , September 2024.
- **Data Sharing-Aware Online Algorithms for Task Allocation in Edge Computing Systems**
IEEE International Conference on Edge Computing, July 2024.
- **Data Sharing-Aware Task Allocation in Edge Computing Systems**

IEEE International Symposium on Women in Service Computing (WISC 2024), July 2024.

Data Sharing-Aware Algorithms for Task Allocation in Edge Computing Systems

IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid 2024), May 2024.

- **Data Sharing-Aware Task Allocation in Edge Computing Systems**

IEEE International Conference on Edge Computing, December 2021.

Workshops

- **Data Sharing-Aware Task Allocation in Edge Computing Systems**

CRA-WP Grad Cohort, Computing Research Association Women Workshop, April 23, 2022.

- **Parallel Fixed-Parameter Tractable Algorithms**

CRA-WP Grad Cohort, Computing Research Association Women Workshop, April 23, 2021.

- **Some Results of the Existence of P_3 - Factor in Regular Graphs**

3rd Istanbul Design Theory, Graph Theory and Combinatorics Workshop, Turkey, June 16-21, 2016.

8th Conference and Workshop on Algebraic Combinatorics and Graph Theory, Iran, April 17, 2015.

University

Parameterized Algorithms

Weekly seminar at Wayne State University, Summer 2019.

Burrows-Wheeler Transform and FM Index

Presentation at University of Central Florida, Summer 2018.

Presentation at Louisiana State University, Fall 2018.

A Rational Approach to the Coloring in Graphs, Fractional Graph Theory

Presentation at Sharif University of Technology, Fall 2013.

Nowhere-Zero Flow in Random Graphs

Presentation at Sharif University of Technology, Fall 2012.

Application of Hyperbolic Geometry in Planar Models of the Hyperbolic Plane “The Poincare Disc Model”

Presentation at Shahid Beheshti University, Summer 2011.

A review on “Gödel Incompleteness Theorem”, Weekly Seminars in Logic

Presentation at Shahid Beheshti University, Fall 2010.

TEACHING

Primary Instructor for three courses at Wayne State University

CSC3110: Algorithm Design and Analysis, Summer 2021 and 2022.

CSC2201: Laboratory for Data Structures and Algorithm Analysis, Winter 2020.

CSC3020: Java Programming, Summer 2019.

Graduate Teaching Assistant for three courses at Wayne State University

CSC6580: Design and Analysis of Algorithms, Fall 2022.

CSC3020: Java Programming, Fall 2019.

CSC4996: Senior Capstone Project, Winter 2019.

Graduate Teaching Assistant for three courses at Louisiana State University

CSC1351: Java Programming, Fall 2017, Spring 2018, Fall 2018.

SERVICE

Reviewer

IEEE Transactions on Automation Science and Engineering (T-ASE 2024).

8th IEEE International Conference on Fog and Edge Computing (ICFEC 2024)

15th IEEE/ACM International Conference on Utility and Cloud Computing (UCC2022)

IEEE Data Compression Conference (DCC 2019)

PROFESSIONAL AFFILIATIONS

ACM (Association for Computing Machinery)

IEEE (Institute of Electrical and Electronics Engineers)

IMS (Iranian Mathematical Society)

IranWiC (Iranian Women in Computer Science)